

Analyzing SSH Log Files Using Splunk SIEM

Objective

To analyze SSH authentication events from system logs using Splunk to identify possible Brute force attacks, unauthorized access attempts, and overall SSH login activity

Tools Used

Splunk Enterprise (Search & Reporting App)

SSH logs File

Dataset Used

Source: ssh_logs.json

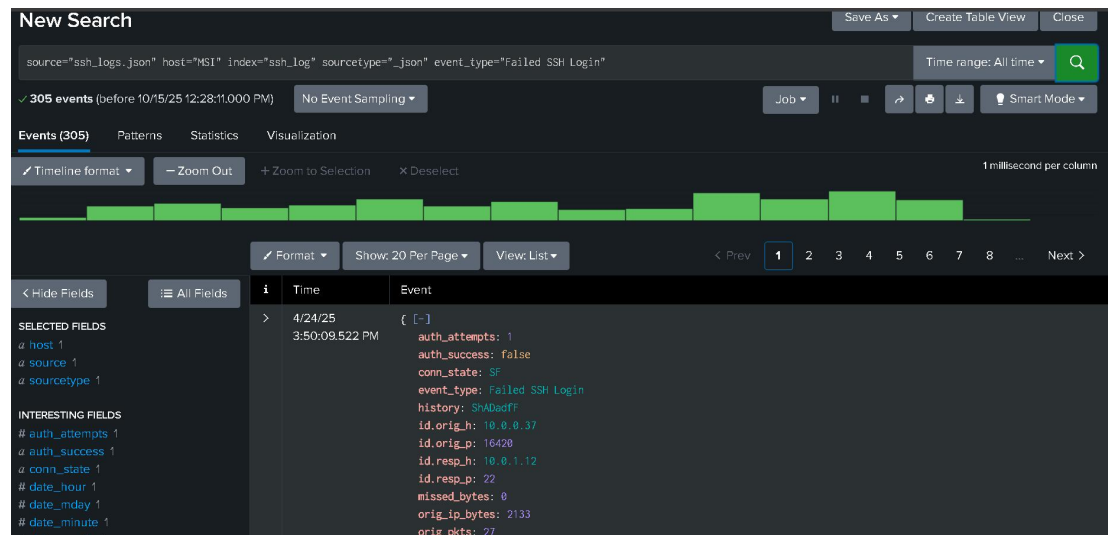
Host: MSI

Sourcetype: _json

Number of total events: 2,400

Queries & Analysis Performed

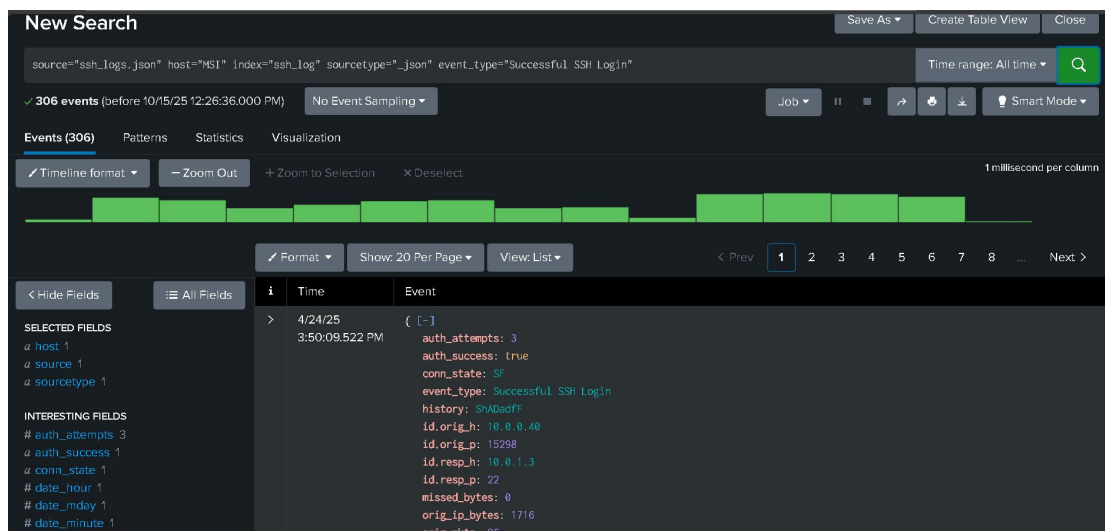
1. Failed SSH Login Attempts



Query

Source="ssh_logs.json" host="MSI" index="ssh_log" sourcetype="_json" event_type="Failed SSH Login"

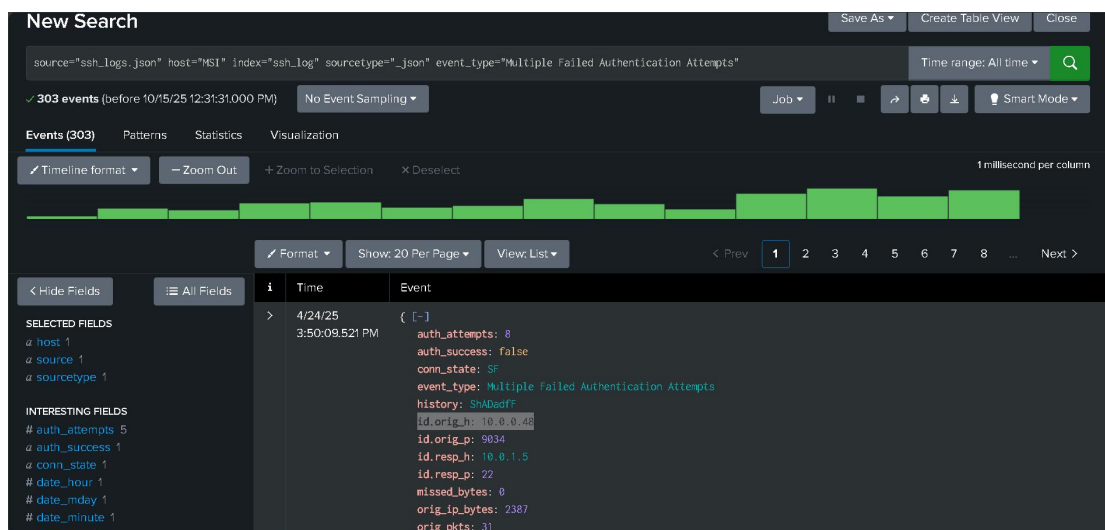
2. Successful login attempt



Query

Source="ssh_logs.json" host="MSI" index="ssh_log" sourcetype="json" event_type="successful SSH Login"

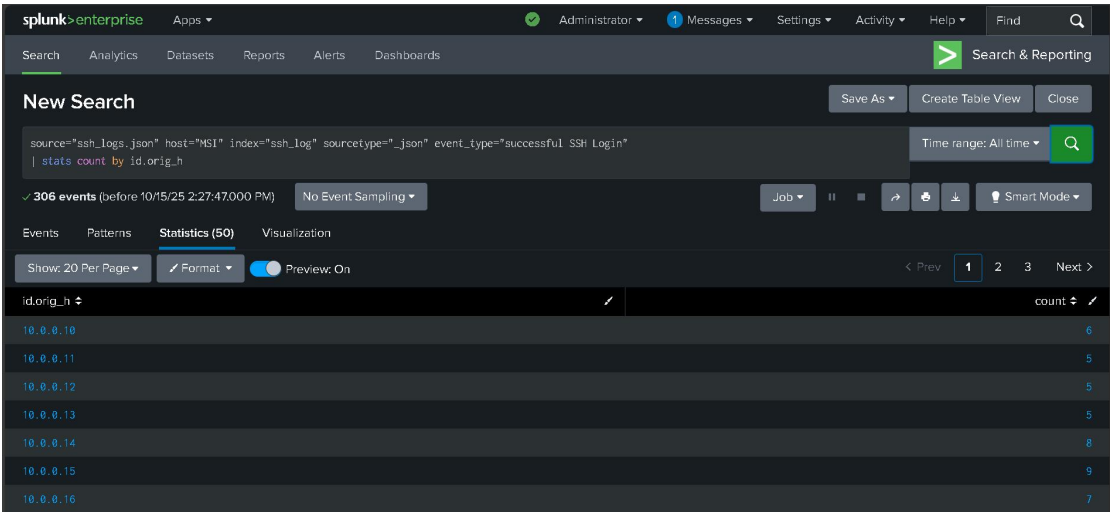
3. ssh multiple failed login attempt



Query

Source="ssh_logs.json" host="MSI" index="ssh_log" sourcetype="json" event_type="multiple failed Authentication Attempt"

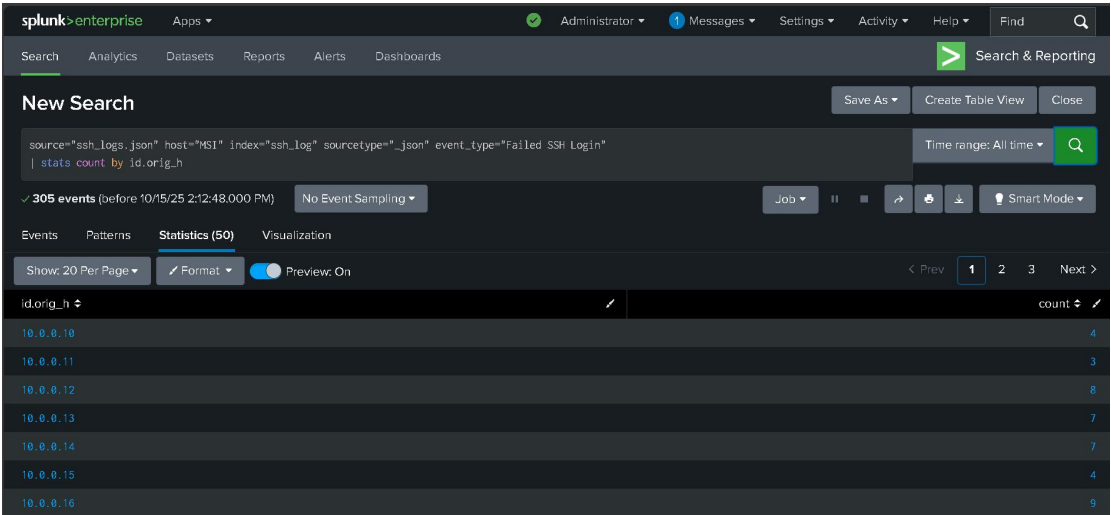
4. Stats count by each source ip of successful login attempt



Query

Source="ssh_logs.json" host="MSI" index=ssh_logs event_type="Successful SSH Login"
| stats count by id.orig_h

5. Stats count by each source ip of Failed login attempt



Query

```
Source="ssh_logs.json" host="MSI" index=ssh_logs event_type="Successful SSH Login"  
| stats count by id.orig_h
```

Suggested Mitigations

- .Implement account lockouts after several failed attempts.
- .Use key based authentication instead of pass words.
- .Enable SSH logging and regularly analyze them using Splunk.
- .Block or restrict access from suspicious IP addresses.

Conclusion

This Splunk based mini project demonstrated how to leverage SIEM tools to analyze SSH log data for identifying malicious behavior patterns and gaining insights into SSH access trends. Such proactive monitoring is essential for maintaining a secure IT environment.